pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Case Study: A Macro Agent on Two Statements

WEEK 2 · DAY 5

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Today: a macro agent on two statements

Week 2, working together

Day 6	market tools over the curve
Day 7	documents as tools, with sources
Day 8	tools behind a server
Day 9	the loop in a framework

Today

1. statement_diff: what changed, no model
2. The macro agent and its MacroNote
3. The answers decided first
4. The note: the step limit, then a fit
5. Every figure twice, every quote twice
6. Why did the curve move?
7. A workbook; what it shows, where it fails
8. Excel, Git, the AI check

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for agents

On the statements: the wording, described; a changed word is a changed word

Allowed names

the statement states

a sentence in one and not the other

the date of the statement

Never called

hawkish or dovish

a policy view

what the Committee will do

On the curve: levels and changes in the file, never a view on rates

Allowed names

the 10 Yr par yield in the file

the change in bp between two dates

the slope in bp

Never called

a forecast or an outlook

a signal

a view on rates

How statement_diff works

The problem	Two statements are mostly the same sentences; what changed needs no model
The method	Normalize both texts, split at a full stop, question mark, or exclamation mark and a space
The rows	Each sentence the other statement does not hold word for word, first side first
You control	The two texts and, in the module, the rule that ends a sentence
What comes out	A table: one row per sentence, and which statement holds it; wording, not meaning

statement_diff: what changed in the wording

2026-07-29

the earlier statement, from the file

2026-09-16

the later one, never typed

```
diff = agentkit.statement_diff(raw[FIRST], raw[SECOND])   # normalize_text first, then split
only in the first: 7; only in the second: 6
```

Why normalize first

The vote line as published holds U+2013, an en dash; normalize_text writes a hyphen with spaces.

Every row is a sentence as published: what the file says, not what it means.

How the splitter cuts

```
a sentence ends at ".", "?", or "!" followed by a space; a line break is a space
```

```
rows that begin with the vote line and run on into the next sentence: 2
rows that end at a single capital letter and a full stop: 2
the same statement with curly apostrophes: 0 rows
```

The vote line

ends with a colon, so it joins the decision

a new vote count changes the joined row

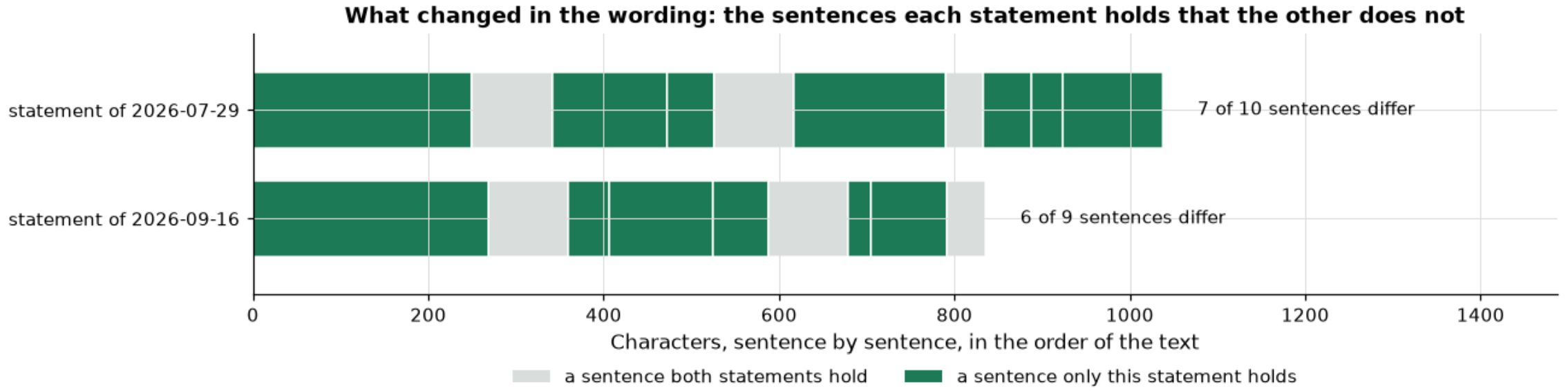
A person's initial

a full stop after it ends a "sentence"

the dissent sentence comes out in three pieces

Any list of changed sentences depends on where its sentences end.

What changed in the wording



10 and 9 sentences, 7 and 6 differing. Where the wording differs, and nothing about why.

How the macro agent works

The agent	A day 9 agent with seven tools and a structured answer, a MacroNote
The run	The model reads the descriptions, requests the statement tools and the curve tools, returns the note
The shape	Quotes for sentences, figures that each cite a tool request; the framework checks only the shape
You control	The tools, the instructions, and the fields of MacroNote
What comes out	A note whose figures can be checked and whose quotes can be searched for

The macro agent and its MacroNote

```
7 tools: ['statement_dates', 'statement_text', 'treasury_yield', 'curve_on', 'curve_change_bp', 'slope_bp', '...
```

```
macro_agent = agentkit.sdk_agent("macro desk", INSTRUCTIONS, list(macro_tools.values()), aclient,
                                 agentkit.MODELS["desk"], 10, output_type=MacroNote)
```

```
output type MacroNote: fields ['text', 'sentences_added', 'sentences_removed', 'figures', 'sources']
each figure: ['name', 'value', 'unit', 'tool_call_id', 'tenor']
store False, prompt_cache_key 'course8-day10'
```

CurveFigure subclasses agentkit.Figure and adds the tenor: check_figures reads a MacroNote as a DeskAnswer.

Every field description goes into the request, so each one is a prompt: no opinion word.

Every setting of the seven tools

Tool	Setting	Type	Default
statement_dates	year	integer	required
statement_text	day	string	required
statement_text	max_chars	integer	required
treasury_yield	tenor	string	required
treasury_yield	as_of	string	required
curve_on	as_of	string	required
curve_change_bp	tenor	string	required
curve_change_bp	start	string	required
curve_change_bp	end	string	required
slope_bp	short	string	required
slope_bp	long	string	required
slope_bp	as_of	string	required
synthetic_spread	kind	string	required
synthetic_spread	as_of	string	required

14 settings, none with a default. Tune first: max_chars of statement_text. statement_diff is not a tool.

The case study: context, objective, data

Context

After each meeting, a desk writes a short note:

what the statement changed, and

what the Treasury curve did.

Objective

A note on the two statements and the curve

Every figure and every quote checked twice

Why did the curve move? Then a workbook

```
statements of 2026: ['2026-01-28', '2026-03-18', '2026-04-29', '2026-06-17', '2026-07-29', '2026-09-16']
2026-07-29 in the curve file: yes
2026-09-16 in the curve file: yes
2026-10-02 in the curve file: yes
2026-09-19 in the curve file: no
```

Course 7 read these statements as data on days 1 and 5. Today looks only at the rows the note touches.

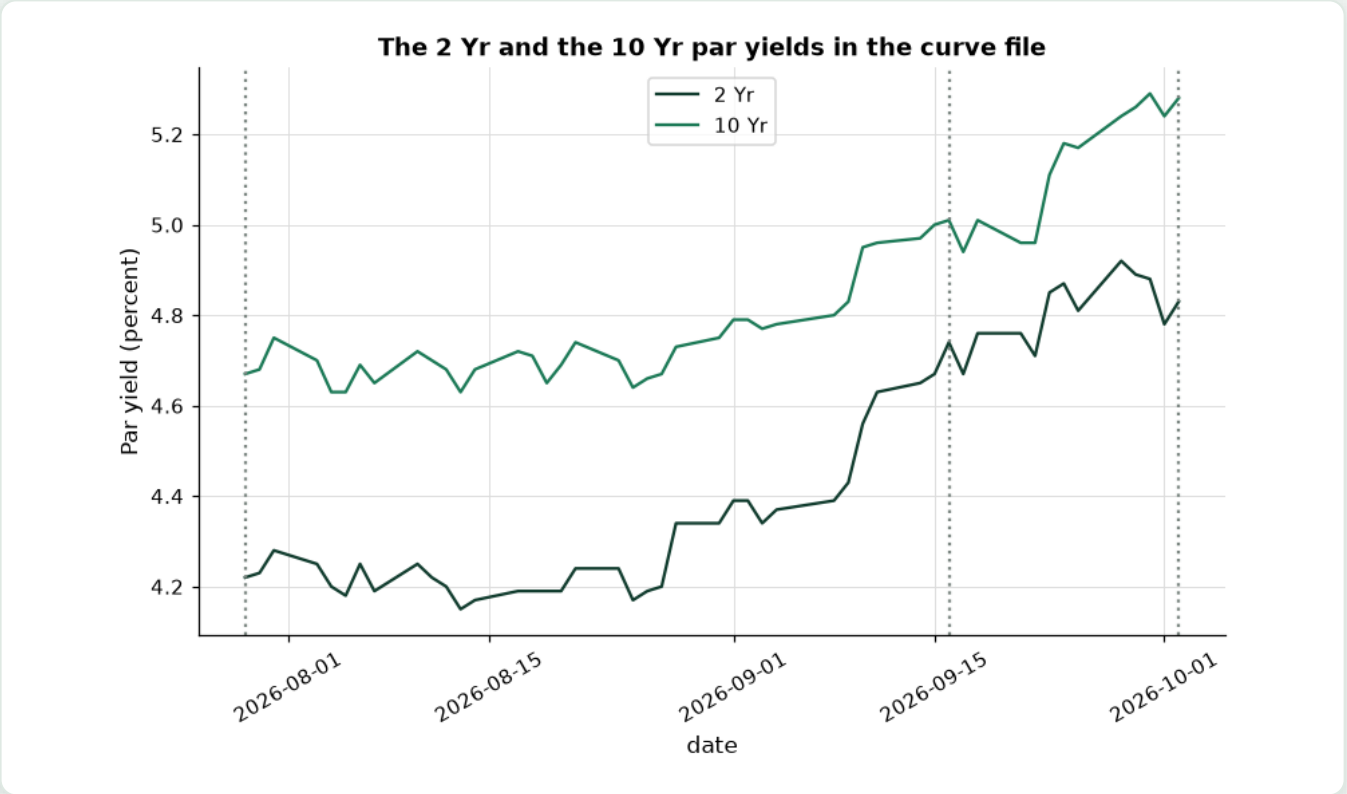
The answers decided first

```
tenor      start        end  start_pct  end_pct  change_bp
 2 Yr 2026-07-29 2026-09-16       4.22     4.74       52.0
 2 Yr 2026-09-16 2026-10-02       4.74     4.83        9.0
10 Yr 2026-07-29 2026-09-16       4.67     5.01       34.0
10 Yr 2026-09-16 2026-10-02       5.01     5.28       27.0
```

```
2 Yr   2026-07-29 to 2026-09-16: the tool equals pandas: yes
2 Yr   2026-09-16 to 2026-10-02: the tool equals pandas: yes
10 Yr  2026-07-29 to 2026-09-16: the tool equals pandas: yes
10 Yr  2026-09-16 to 2026-10-02: the tool equals pandas: yes
```

What the file holds, decided before any request. No file of the course says why either yield moved.

The two yields and the three dates



The chart describes the file. It says nothing about why a line moved.

The first form: the step limit

8

tool requests, then the limit

no

note returned

```
the run stopped at the step limit: Max turns (8) exceeded
tool requests: 8; {'statement_text': 2, 'treasury_yield': 6}
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 8 requests, 17,671 prompt and 224 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

Ten numbers one by one, plus two statements: twelve requests. curve_change_bp carries all ten in four.

The fix is not a larger limit. It is a question that names the tool and counts the requests.

The note, and its quotes against statement_diff

5
sentences added

5
removed

10
figures

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 7 requests, 15,322 prompt and 1,407 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The note's quotes

every one in its statement, word for word

7 are rows of statement_diff

3 are cut where the diff did not cut

statement_diff's rows

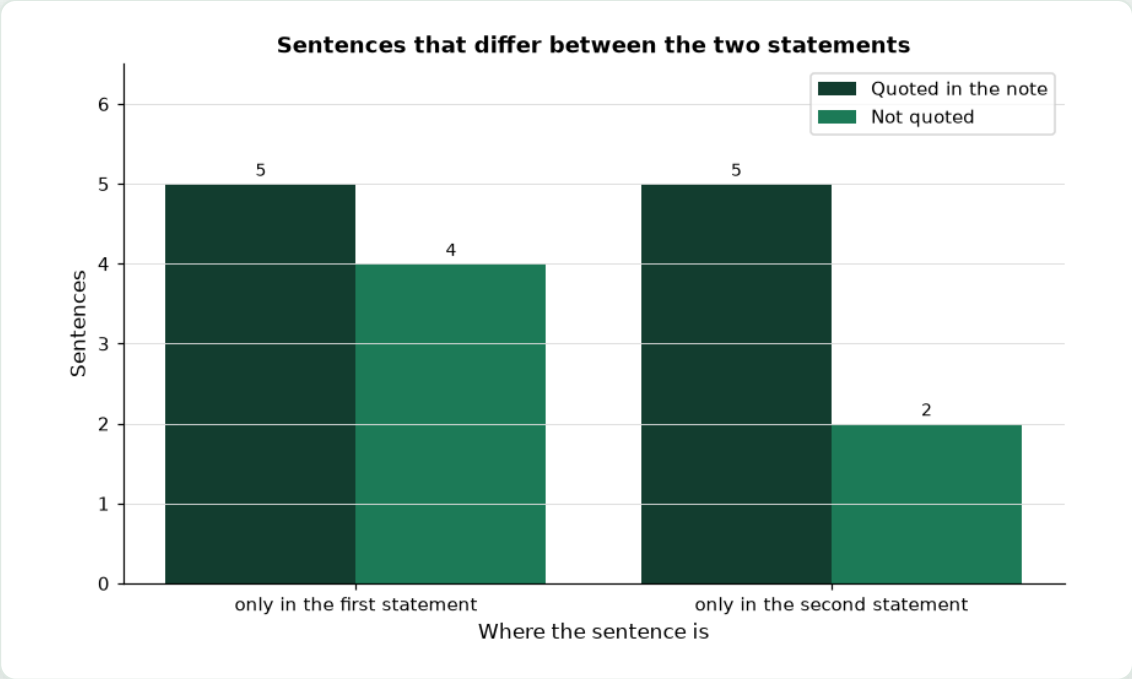
7 quoted by the note, 6 not

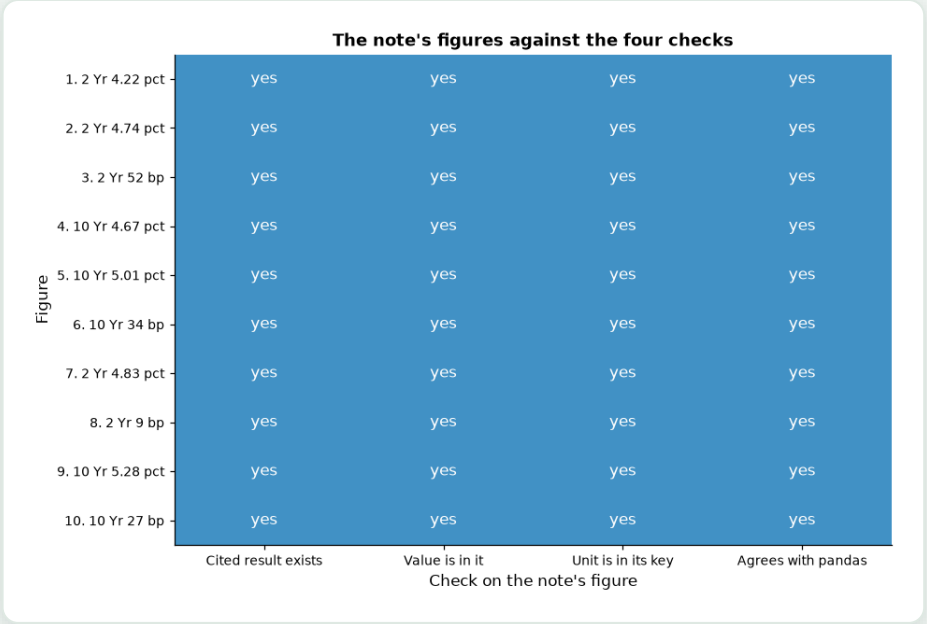
5 of those: the same text, cut another way

1 left out: "Inflation remains elevated."

Two faithful lists can disagree. Check: is it in the statement word for word? Is it a change?

The quotes and the figures, drawn





Left: which differing sentences the note quotes. Right: every figure against four checks.

The charts are the tables above, drawn. A light cell names a figure to look at.

How the comparison works

The tests	Three, each asking a different question of the note
Figures	check_figures follows the citation; agrees compares with pandas' number for the same tenor
Quotes	Searched for in the statements and in statement_diff's rows
You control	The key and the rounding
What comes out	A row per figure, a row per sentence, and a verdict on the note

Every figure twice, and the note's own words

```
figures: 10; traced: 10; agree with pandas: 10; key numbers stated: 12 of 12
```

```
numbers in the note's text that no tool returned: []
```

```
assert_extraction_only: the text uses forecast, advice, or lending words: ['will']
```

Where the word came from

The note's own text restated a sentence of the later statement, in its own voice, without quotes.

The check cannot tell a restated sentence from a forecast; the note quotes, and describes.

A changed word is a changed word: the note may quote it, and never says what the Committee will do.

How a why question works

The test	What the tools can answer: the same seven tools, a text answer
What exists	What each statement says, and how far a yield moved; no tool returns a reason
The answer	The numbers, and NOT IN TEXT for the reason
You control	The question and the instructions, which ask for a source for every claim
What comes out	Traced numbers; the explanation, if any, is the part to read

Why did the curve move?

```
WHY_QUESTION: by how many bp did the 10 Yr par yield change between the two dates, and why?
```

The number: from the tool

34 bp, from 4.67 to 5.01 percent

curve_change_bp's result, traced

The link: from no file

"likely contributed": the agent's sentence

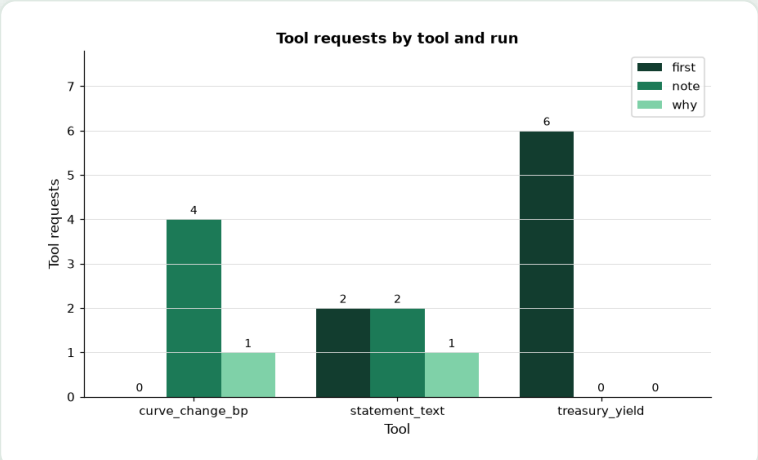
the files hold no reason for a move

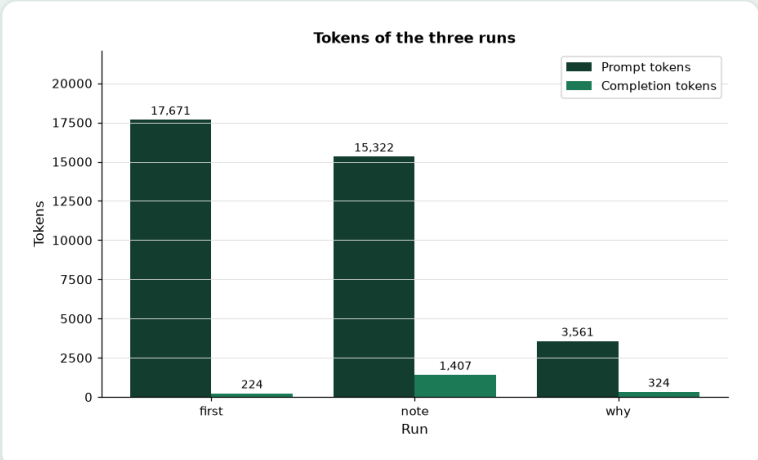
Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 3,561 prompt and 324 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

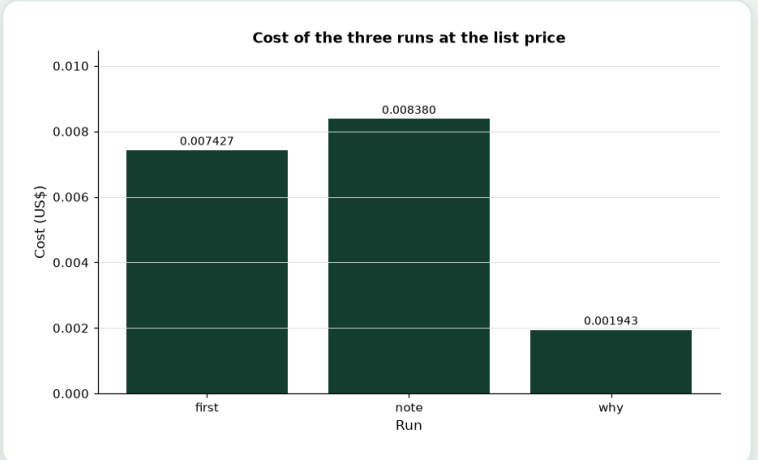
```
numbers in the answer that no tool returned: []
```

Every number traces; the reason does not. The checks catch numbers. A reader catches reasons.

The three runs, side by side







Run	Requests	Tool requests	Prompt	Completion	Cost (US$)
first	8	8	17,671	224	0.007427
note	7	6	15,322	1,407	0.008380
why	3	2	3,561	324	0.001943

What the agent shows, and where it fails

```
sheets: ['figures', 'sentences']; rows [10, 16]
```

Shows	10 figures traced and agreeing; 10 quotes word for word
Fails: the limit	the first form listed ten numbers; no note
Fails: the lists	other sentence ends; the inflation line left out
Fails: own words	the Committee's word restated in the note's voice
Fails: why	a link to the decision that no file states

The curve moved between the dates; no file says why, and a note may not either.

Excel side by side, and git tag week2

```
=(INDEX(B2:O2941,MATCH(DATE(2026,9,16),A2:A2941,0),MATCH("2 Yr",B1:O1,0))
  -INDEX(B2:O2941,MATCH(DATE(2026,7,29),A2:A2941,0),MATCH("2 Yr",B1:O1,0)))*100
```

2 Yr change, 2026-07-29 to 2026-09-16	`Excel 52`
2 Yr change, 2026-09-16 to 2026-10-02	`Excel 9`
10 Yr change, 2026-07-29 to 2026-09-16	`Excel 34`
10 Yr change, 2026-09-16 to 2026-10-02	`Excel 27`
10 Yr change, 2026-07-29 to 2026-09-19	`Excel #N/A`
10 Yr, 2026-07-29 to 2026-09-16, without x 100	`Excel 0.34`

```
week2
(HEAD -> main, tag: week2) agentkit day 10: statement_diff, 52 tests
```

Excel 16.0 build 20430, typed by script; the workbook read back equal. 52 passed; the tag week2.

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

The later yield less the earlier, times 100: 1 percent is 100 bp.

2

Code back

Written for the course in the style of an AI answer. It runs.

3

Test first

Two hand rows, 4.00 and 4.25 percent. It returns 0.25 and fails.

4

Fix

The difference times 100: 25.0 on the rows, 34.0 on the file.

Today: curve_change_bp returns the change in percent and labels it bp: a 25 bp move prints as 0.25.

Code that runs is not the same as code that is right.

What you can do now

- statement_diff: the changed sentences, and how it cuts them
- A framework agent whose answer is your own MacroNote
- A question that fits the tools and the step limit
- Every figure and every quote checked twice
- No file says why the curve moved; read for that

Week 3, Day 11 An orchestrator and specialists